

Non-Functional Requirements: Virtual Classroom Platform

Usability:

Description: A first-time user should be able to navigate the platform and join a virtual class session within five minutes without additional help.

Metric: 90% of users should be able to complete their initial tasks (e.g., joining a class, submitting an assignment) without assistance after their first login.

Priority Level: High

Precondition: User has access to a compatible device with a stable internet connection.

Performance:

Description: The platform shall support at least 500 concurrent users per session without noticeable degradation in video/audio quality.

Metric: Response time for accessing any page or feature should be under two seconds under normal load and a maximum of four seconds under peak load.

Priority Level: High

Precondition: Server infrastructure and internet bandwidth meet minimum requirements for load handling.

Reliability:

Description: The system should be available 99.5% of the time, excluding scheduled maintenance.

Metric: System downtime should not exceed 4 hours per month.

Priority Level: High

Precondition: Maintenance schedules and disaster recovery plans are implemented.

Scalability:

Description: The platform shall be able to scale to support an increase in users (up to 10,000 concurrent users) without requiring significant redesign.

Metric: System performance (page load times, video quality) should be maintained within acceptable limits as user count increases.

Priority Level: Medium

Precondition: Scalable architecture and cloud-based resources are available.

Security:

Description: All user data, including login credentials and personal information, shall be securely encrypted during transmission and storage.

Metric: The platform shall comply with industry-standard encryption protocols (e.g., SSL/TLS) and ensure data breaches are minimized to a risk level of less than 1%.

Priority Level: High

Precondition: Security policies are implemented and encryption mechanisms are in place.

Compatibility:

Description: The platform should be accessible on major web browsers (e.g., Chrome, Firefox, Safari) and be compatible with mobile devices (iOS and Android).

Metric: The platform should maintain full functionality on 90% of devices and browsers currently in common use.

Priority Level: Medium

Precondition: The latest versions of major browsers and operating systems are supported.

Maintainability:

Description: The platform codebase should be modular to allow future updates and changes without affecting existing functionality.

Metric: Updates and patches should be deployable with minimal downtime (less than 1 hour) and require no significant code rewrites.

Priority Level: Medium

Precondition: Codebase follows modular design principles and is well-documented.

Accessibility:

Description: The platform shall be compliant with accessibility standards (e.g., WCAG 2.1) to support users with disabilities.

Metric: 95% of all key features (classroom access, assignment submission) should be accessible via screen readers and keyboard navigation.

Priority Level: High

Precondition: Accessibility testing and inclusive design standards are integrated into development.

Project Development Constraints: Virtual Classroom Platform

Budget Constraints

- **Description:** The project must stay within a fixed budget, limiting available resources, tools, and technologies. This may restrict the ability to purchase premium software, hire additional developers, or implement advanced features.
- **Priority Level:** High
- **Precondition:** Initial budget assessment and allocation for essential features, resources, and tools.

Time Constraints

- **Description:** The platform must be developed and launched within a set timeline to meet educational institution schedules (e.g., the start of an academic semester). This may require prioritizing certain features over others to meet deadlines.
- **Priority Level:** High
- **Precondition:** Project plan and timeline are established, with key milestones defined for each development phase.

Technology Constraints

- **Description:** The platform must be compatible with existing infrastructure, such as specific operating systems, browsers, or hardware used by the educational institution. This may limit the choice of technologies and frameworks that can be used for development.
- **Priority Level:** High
- **Precondition:** Assessment of educational institution's existing technology environment and compatibility requirements.

Compliance Constraints

- **Description:** The platform must adhere to industry regulations and standards, such as FERPA (Family Educational Rights and Privacy Act) for student data privacy in the U.S., or GDPR (General Data Protection Regulation) for data protection if serving users in the EU.
- **Priority Level:** High
- **Precondition:** Identification of applicable regulations and requirements for compliance, with legal counsel or compliance experts consulted as necessary.

Resource Constraints

- **Description:** The project team may be limited in terms of available personnel, skills, and expertise. For instance, if there are limited UI/UX designers or cybersecurity experts, certain aspects of the platform may be constrained by available talent.
- **Priority Level:** Medium
- **Precondition:** Evaluation of team skills and identification of gaps, with potential recruitment or upskilling options explored as needed.

Scalability and Infrastructure Constraints

- **Description:** The platform must operate within the available server capacity or cloud infrastructure, which may limit the number of concurrent users and restrict certain features, such as real-time video conferencing.
- **Priority Level:** Medium
- **Precondition:** Review of existing server or cloud infrastructure to assess load-handling capacity and scalability.

Security Constraints

- **Description:** To ensure user data protection, the platform must follow stringent security protocols and encryption standards. This may require additional development time and restrict the use of certain third-party tools or plugins.
- **Priority Level:** High
- **Precondition:** Establishment of security policies, with protocols defined for data encryption, secure data storage, and regular security audits.